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3627 Lines (3627 loc) · 460 KB

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Title: DSC 680: Project 1: Creating Thirs Shot Strategy in Pickleball

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Date: 31 October 2025

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Description: This project reads in cleans and gathersa the essential data to train and evaluate predictive model predicting the proper shot selection to win a rally. There will be two models train one more complex for professional players, one less complex for amateur players.

Project 2: Shotselection in Pickleball

Connect to the PKLMart Database and collect data.

```
In [73]: # Connect to the pklmart database to confirm access
import psycopg2

conn = psycopg2.connect(
    host="pklmartdb.ckkvdwandqoa.us-east-1.rds.amazonaws.com",
    port=5432,
    dbname="postgres",
    user="mheinen",
    password="4A3FR21l(!"
)
cur = conn.cursor()
cur.execute("SELECT version();")
print(cur.fetchone())
cur.close()
conn.close()
```

('PostgreSQL 13.20 on x86_64-pc-linux-gnu, compiled by gcc (GCC) 7.3.1 20180712 (Red Hat 7.3.1-17), 64-bit',)

```
In [74]: # Show all the tables in the database
# only pklm.prd schema,
conn = psycopg2.connect(
    host="pklmartdb.ckkvdwandqoa.us-east-1.rds.amazonaws.com",
    port=5432,
    dbname="postgres",
    user="mheinen",
    password="4A3FR21l(!"
)
cur = conn.cursor()
cur.execute("""
    SELECT table_schema, table_name
    FROM information_schema.tables
    WHERE table_type = 'BASE TABLE' AND table_schema = 'pklm_prd';
""")
tables = cur.fetchall()
for schema, table in tables:
    print(f'{schema}.{table}')
cur.close()
conn.close()
```

pklm_prd.ball_type_ref
pklm_prd.metrics_ref
pklm_prd.shot_temp
pklm_prd.player_dupr_hist
pklm_prd.shot
pklm_prd.cv_player_pos
pklm_prd.rally
pklm_prd.backfill
pklm_prd.match
pklm_prd.game
pklm_prd.team
pklm_prd.player
pklm_prd.model_info
pklm_prd.tournament
pklm_prd.baseline_metrics
pklm_prd.game_temp
pklm_prd.up_player_cw
pklm_prd.player_dupr
pklm_prd.shot_type_ref

```
In [75]: # Read in the rally data from the database
import pandas as pd
# Establish a connection to the PostgreSQL database
conn = psycopg2.connect(
    host="pklmartdb.ckkvdwandqoa.us-east-1.rds.amazonaws.com",
    port=5432,
    dbname="postgres",
    user="mheinen",
    password="4A3FR21l(!"
)
query = "SELECT * FROM pklm_prd.rally;"
rally = pd.read_sql(query, conn)
conn.close()

# Preview the rally data
rally.head(5)
```

/var/folders/xv/q0156cqs67s40wbxq6_vfykr0000gn/T/ipykernel_19613/3773201166.py:12: UserWarning: pandas only support s SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 obje cts are not tested. Please consider using SQLAlchemy.

```
Out[75]: rally_id match_id game_id rally_nbr srv_team_id srv_player_id rtn_team_id rtn_player_id ts_player_id ts_type ... rtrr
```

0	R8968	M113	G208	6	T152	P202	T155	P70	P202	Drop	...
1	R8963	M113	G208	1	T152	P202	T155	P70	N/A	Drop	...
2	R8964	M113	G208	2	T155	P70	T152	P202	P205	Drive	...
3	R8965	M113	G208	3	T155	P205	T152	P203	P205	Drop	...

```
4      R8966      M113      G208      4      T155      P205      T152      P202      P70      Drop      ...
```

5 rows x 30 columns

```
In [76]: #Show the shape of the rally data
rally.shape
```

Out[76]: (54058, 30)

```
In [77]: # Show all the columns in the rally data
rally.columns
```

Out[77]: Index(['rally_id', 'match_id', 'game_id', 'rally_nbr', 'srv_team_id', 'srv_player_id', 'rtrn_team_id', 'rtrn_player_id', 'ts_player_id', 'ts_type', 'w_team_id', 'to_ind', 'to_team_id', 'rally_len', 'srv_switch_ind', 'rtrn_switch_ind', 'srv_team_flipped_ind', 'rtrn_team_flipped_ind', 'srv_team_rs_player_id', 'srv_team_ls_player_id', 'rtrn_team_rs_player_id', 'rtrn_team_ls_player_id', 'ending_type', 'ending_player_id', 'lob_cnt', 'dink_cnt', 'maint_dtm', 'maint_app', 'create_dtm', 'create_app'], dtype='object')

```
In [78]: # Remove uneseccsary or irrelevant columns from the rally data
columns_to_keep = ['rally_id', 'rally_nbr', 'srv_team_id', 'srv_player_id', 'rtrn_team_id', 'rtrn_player_id', 'ts_player_id', 'ts_type', 'w_team_id', 'rally_len', 'srv_switch_ind', 'rtrn_switch_ind', 'srv_team_flipped_ind', 'rtrn_team_flipped_ind', 'srv_team_rs_player_id', 'srv_team_ls_player_id', 'rtrn_team_ls_player_id']
rally = rally[columns_to_keep]
# Preview the filtered DF
rally.head(5)
```

Out[78]:

	rally_id	rally_nbr	srv_team_id	srv_player_id	rtrn_team_id	rtrn_player_id	ts_player_id	ts_type	w_team_id	rally_len	srv_sv
0	R8968	6	T152	P202	T155	P70	P202	Drop	T152	14.0	
1	R8963	1	T152	P202	T155	P70	N/A	Drop	T155	13.0	
2	R8964	2	T155	P70	T152	P202	P205	Drive	T152	5.0	
3	R8965	3	T155	P205	T152	P203	P205	Drop	T155	8.0	
4	R8966	4	T155	P205	T152	P202	P70	Drop	T155	11.0	

```
In [79]: # shot level data to get the location of the returns for all thirs shots to help train the model
# Establish a connection to the PostgreSQL database
conn = psycopg2.connect(
    host="pklmartdb.ckkvdwandqoa.us-east-1.rds.amazonaws.com",
    port=5432,
    dbname="postgres",
    user="mheinen",
    password="4A3FR211(!"
)
query = """
SELECT shot_id, rally_id, loc_x, loc_y, btt_before, shot_type, shot_type_orig
FROM pklm_prd.shot
WHERE shot_nbr = 3;
"""
shot_third = pd.read_sql(query, conn)
conn.close()
# Preview the third shot data
shot_third.head(5)
```

/var/folders/xv/q0156cqs67s40wbxq6_vfykr0000gn/T/ipykernel_19613/3480373998.py:15: UserWarning: pandas only support s SQLAlchemy connectable (engine/connection) or database string URI or sqlalchemy DBAPI2 connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.

shot_third = pd.read_sql(query, conn)

Out[79]:

	shot_id	rally_id	loc_x	loc_y	btt_before	shot_type	shot_type_orig
0	S150129	R16839	12.30	19.75	0.4649	None	tsDrv
1	S150084	R16837	4.67	17.09	0.3325	None	tsDrp
2	S150102	R16838	2.92	23.24	0.5436	None	tsDrp
3	S436638	R52813	1.73	25.35	0.3583	None	tsDrv
4	S150146	R16841	2.66	21.68	0.3615	None	tsDrp

```
In [80]: # find the percentage of missing values in the third shot data
missing_percentage = shot_third.isnull().mean() * 100
missing_percentage
```

Out[80]: shot_id 0.000000
rally_id 0.000000
loc_x 1.388625
loc_y 1.401269
btt_before 0.891333
shot_type 98.539731
shot_type_orig 0.000000
dtype: float64

```
In [81]: # Remove shot_type from the third shot data due to high missing values
shot_third = shot_third.drop(columns=['shot_type'])

# Apply the average loc_x and loc_y and btt_before for missing values
shot_third['loc_x'].fillna(shot_third['loc_x'].mean(), inplace=True)
shot_third['loc_y'].fillna(shot_third['loc_y'].mean(), inplace=True)
shot_third['btt_before'].fillna(shot_third['btt_before'].mean(), inplace=True)

# find the percentage of missing values in the third shot data
missing_percentage = shot_third.isnull().mean() * 100
missing_percentage
```

Out[81]: shot_id 0.0
rally_id 0.0
loc_x 0.0

```
loc_y      0.0
btt_before 0.0
shot_type_orig 0.0
dtype: float64
```

```
In [82]: # Drop shot_id from the third shot data as it is not needed for the merge
shot_third = shot_third.drop(columns=['shot_id'])
```

```
In [83]: # find player specific data to train the model for relevancy against an opponent
# Establish a connection to the PostgreSQL database
conn = psycopg2.connect(
    host="pklmartdb.ckkvdwandqoa.us-east-1.rds.amazonaws.com",
    port=5432,
    dbname="postgres",
    user="mheinen",
    password="4A3FR21l(!"
)
query = """
SELECT player_id, gender, dom_hand
FROM pklm_prd.player;
"""
player = pd.read_sql(query, conn)
conn.close()
# Preview the player data shape
player.shape
```

```
/var/folders/xv/q0156cqs67s40wbxq6_vfykr0000gn/T/ipykernel_19613/1958867211.py:14: UserWarning: pandas only support
s SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 obje
cts are not tested. Please consider using SQLAlchemy.
player = pd.read_sql(query, conn)
```

```
Out[83]: (1077, 3)
```

```
In [84]: # Use a vlookup type merge to get the player data into the rally df for each player
for col in ['srv_team_rs_player_id', 'srv_team_ls_player_id', 'rtrn_team_rs_player_id', 'rtrn_team_ls_player_id']:
    rally = rally.merge(
        player.rename(columns={'player_id': col, 'gender': f'{col}_gender', 'dom_hand': f'{col}_dom_hand'}),
        on=col,
        how='left'
    )
# Preview the merged DataFrame
rally.head(5)
```

```
Out[84]:
```

	rally_id	rally_nbr	srv_team_id	srv_player_id	rtrn_team_id	rtrn_player_id	ts_player_id	ts_type	w_team_id	rally_len	...	rt
0	R8968	6	T152	P202	T155	P70	P202	Drop	T152	14.0	...	
1	R8963	1	T152	P202	T155	P70	N/A	Drop	T155	13.0	...	
2	R8964	2	T155	P70	T152	P202	P205	Drive	T152	5.0	...	
3	R8965	3	T155	P205	T152	P203	P205	Drop	T155	8.0	...	
4	R8966	4	T155	P205	T152	P202	P70	Drop	T155	11.0	...	

5 rows x 26 columns

```
In [85]: # Merge the shot third data with the rally data to get the loc_x and loc_y, btt_before for the third shot
# use a left join to keep all rally data
rally_ts = rally.merge(
    shot_third,
    on='rally_id',
    how='left'
)
# Preview the merged DataFrame
rally_ts.head(5)
```

```
Out[85]:
```

	rally_id	rally_nbr	srv_team_id	srv_player_id	rtrn_team_id	rtrn_player_id	ts_player_id	ts_type	w_team_id	rally_len	...	sr
0	R8968	6	T152	P202	T155	P70	P202	Drop	T152	14.0	...	
1	R8963	1	T152	P202	T155	P70	N/A	Drop	T155	13.0	...	
2	R8964	2	T155	P70	T152	P202	P205	Drive	T152	5.0	...	
3	R8965	3	T155	P205	T152	P203	P205	Drop	T155	8.0	...	
4	R8966	4	T155	P205	T152	P202	P70	Drop	T155	11.0	...	

5 rows x 30 columns

```
In [86]: # find all the distinct values for the ts_type and shot_type_orig columns
rally_ts['ts_type'].unique(), rally_ts['shot_type_orig'].unique()
```

```
Out[86]: (array(['Drop', 'Drive', 'N/A', '', 'Lob'], dtype=object),
array(['tsDrp', 'tsDrv', nan, 'tsLob', '0', ''], dtype=object))
```

```
In [87]: # change the values of shot type_orig to be more descriptive
# change the values of shot_type_orig to be more descriptive
rally_ts['shot_type_orig'] = rally_ts['shot_type_orig'].replace({'tsDrp': 'Drop', 'tsDrv': 'Drive', 'tsLob': 'Lob'})

# Calculate the number of records where ts_type and shot_type_orig different.
mismatch_count = (rally_ts['ts_type'] != rally_ts['shot_type_orig']).sum()
mismatch_count
```

```
Out[87]: 6675
```

```
In [88]: # find the shape of rally_ts
rally_ts.shape
```

```
Out[88]: (54058, 30)
```

```
In [89]: # Remove all records where ts_type and shot_type_orig are different
rally_ts = rally_ts[rally_ts['ts_type'] == rally_ts['shot_type_orig']]

# Drop columns that are '' for ts_type and shot_type_orig
rally_ts = rally_ts[(rally_ts['ts_type'] != '') & (rally_ts['shot_type_orig'] != '')]
```

```
# find all the distinct values for the ts_type and shot_type_orig columns again to confirm
rally_ts['ts_type'].unique(), rally_ts['shot_type_orig'].unique()
```

Out[89]: (array(['Drop', 'Drive', 'Lob'], dtype=object),
array(['Drop', 'Drive', 'Lob'], dtype=object))

```
In [90]: # Drop the shot_type_orig column as it is now redundant
rally_ts = rally_ts.drop(columns=['shot_type_orig'])
```

```
In [91]: # Preview columns in rally_ts
rally_ts.columns
```

Out[91]: Index(['rally_id', 'rally_nbr', 'srv_team_id', 'srv_player_id', 'rtrn_team_id',
'rtrn_player_id', 'ts_player_id', 'ts_type', 'w_team_id', 'rally_len',
'srv_switch_ind', 'rtrn_switch_ind', 'srv_team_flipped_ind',
'rtrn_team_flipped_ind', 'srv_team_rs_player_id',
'srv_team_ls_player_id', 'rtrn_team_rs_player_id',
'rtrn_team_ls_player_id', 'srv_team_rs_player_id_gender',
'srv_team_ls_player_id_dom_hand', 'srv_team_ls_player_id_gender',
'srv_team_ls_player_id_dom_hand', 'rtrn_team_rs_player_id_gender',
'rtrn_team_rs_player_id_dom_hand', 'rtrn_team_ls_player_id_gender',
'rtrn_team_ls_player_id_dom_hand', 'loc_x', 'loc_y', 'btt_before'],
dtype='object')

```
In [92]: # Feature engineer if the serving team won (ie if third shot was Classify the rally as a win and loss for the serv
# If w_team_id is the same as srv_team_id then serving team won
rally_ts['srv_team_won'] = (rally_ts['w_team_id'] == rally_ts['srv_team_id']).astype(int)

# Validate the result
rally_ts[['srv_team_id', 'w_team_id', 'srv_team_won']].head(10)
```

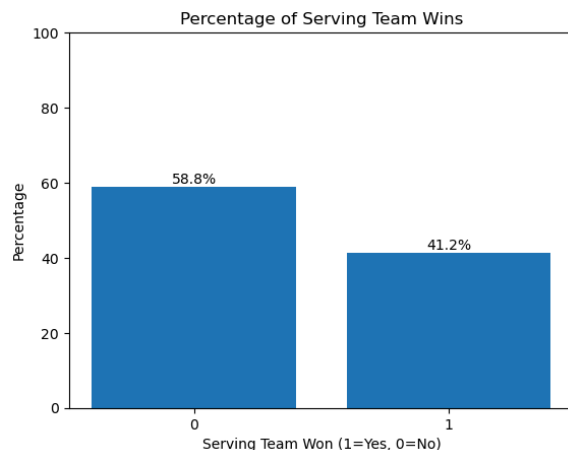
Out[92]:

	srv_team_id	w_team_id	srv_team_won
0	T152	T152	1
1	T152	T155	0
2	T155	T152	0
3	T155	T155	1
4	T155	T155	1
5	T155	T152	0
6	T152	T155	0
7	T152	T155	0
8	T155	T152	0
9	T155	T155	1

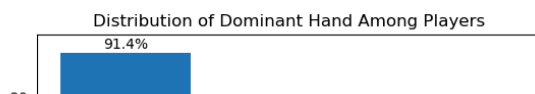
Do some EDA to learn some distributions of the df.

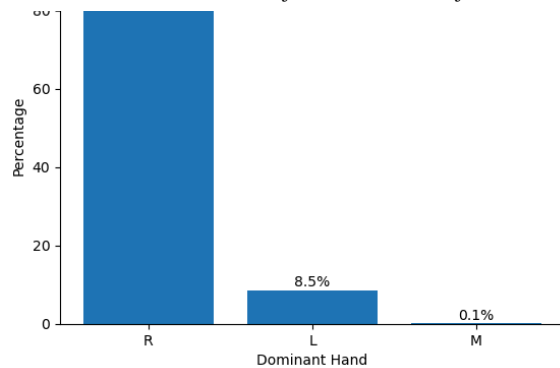
```
In [93]: import matplotlib.pyplot as plt

percentages = rally_ts['srv_team_won'].value_counts(normalize=True) * 100
plt.bar(percentages.index.astype(str), percentages.values)
plt.title('Percentage of Serving Team Wins')
plt.xlabel('Serving Team Won (1=Yes, 0=No)')
plt.ylabel('Percentage')
for i, v in enumerate(percentages.values):
    plt.text(i, v + 1, f'{v:.1f}%', ha='center')
plt.ylim(0, 100)
plt.show()
```



```
In [94]: # Find the percentages of left and right handed players from the player df
# Build a bar chart to show the distribution of dominant hand among players
dom_hand_counts = player['dom_hand'].value_counts(normalize=True) * 100
plt.bar(dom_hand_counts.index, dom_hand_counts.values)
plt.title('Distribution of Dominant Hand Among Players')
plt.xlabel('Dominant Hand')
plt.ylabel('Percentage')
for i, v in enumerate(dom_hand_counts.values):
    plt.text(i, v + 1, f'{v:.1f}%', ha='center')
plt.show()
```





Begin training the models

```
In [95]: # Rename the rally_ts dataframe to model_df for clarity
model_df = rally_ts.copy()
```

```
In [96]: # Preview the columns in model_df
model_df.columns
```

```
Out[96]: Index(['rally_id', 'rally_nbr', 'srv_team_id', 'srv_player_id', 'rtrn_team_id',
'rtrn_player_id', 'ts_player_id', 'ts_type', 'w_team_id', 'rally_len',
'srv_switch_ind', 'rtrn_switch_ind', 'srv_team_flipped_ind',
'rtrn_team_flipped_ind', 'srv_team_rs_player_id',
'srv_team_ls_player_id', 'rtrn_team_rs_player_id',
'rtrn_team_ls_player_id', 'srv_team_rs_player_id_gender',
'srv_team_rs_player_id_dom_hand', 'srv_team_ls_player_id_gender',
'srv_team_ls_player_id_dom_hand', 'rtrn_team_rs_player_id_gender',
'rtrn_team_rs_player_id_dom_hand', 'rtrn_team_ls_player_id_gender',
'rtrn_team_ls_player_id_dom_hand', 'loc_x', 'loc_y', 'btt_before',
'srv_team_won'],
dtype='object')
```

In order to evaluate which shot should be selected we will filter to only include where the serving team wins and then predict the `ts_type`. That leads to that win. This model then then implements in simulation for a team to develop a strategy around.

```
In [97]: #Drop all records with missing values from the model_df
model_df = model_df.dropna()
```

```
# find missing values that are 'NaN'
missing_percentage = model_df.isnull().mean() * 100
missing_percentage
```

```
Out[97]: rally_id          0.0
rally_nbr          0.0
srv_team_id        0.0
srv_player_id       0.0
rtrn_team_id        0.0
rtrn_player_id      0.0
ts_player_id        0.0
ts_type            0.0
w_team_id           0.0
rally_len           0.0
srv_switch_ind       0.0
rtrn_switch_ind      0.0
srv_team_flipped_ind 0.0
rtrn_team_flipped_ind 0.0
srv_team_rs_player_id 0.0
srv_team_ls_player_id 0.0
rtrn_team_rs_player_id 0.0
rtrn_team_ls_player_id 0.0
srv_team_rs_player_id_gender 0.0
srv_team_rs_player_id_dom_hand 0.0
srv_team_ls_player_id_gender 0.0
srv_team_ls_player_id_dom_hand 0.0
rtrn_team_rs_player_id_gender 0.0
rtrn_team_rs_player_id_dom_hand 0.0
rtrn_team_ls_player_id_gender 0.0
rtrn_team_ls_player_id_dom_hand 0.0
loc_x              0.0
loc_y              0.0
btt_before         0.0
srv_team_won        0.0
dtype: float64
```

```
In [98]: # Remove rally length column because it is not realistic to include in the model.
model_df = model_df.drop(columns=['rally_len'])

# show the shape of the model_df
model_df.shape
```

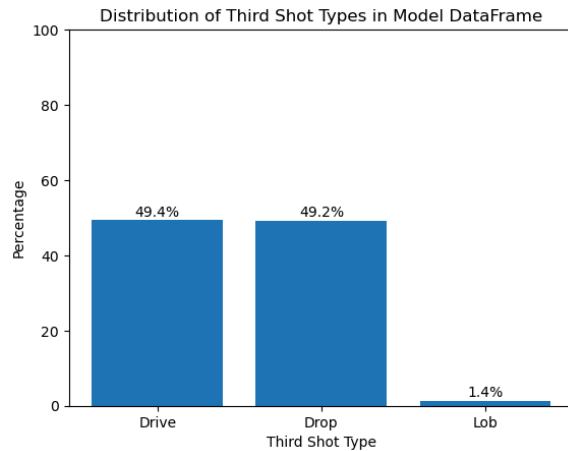
```
Out[98]: (47259, 29)
```

```
In [99]: # Drop the record if there is a column value = 'NaN'
model_df = model_df.replace('NaN', pd.NA).dropna()
```

```
In [100]: # Filter the model_df to only include winning serving team records
model_df = model_df[model_df['srv_team_won'] == 1]
```

```
In [120]: # Build a bar graph showing the distribution of ts_type in the model_df
ts_type_counts = model_df['ts_type'].value_counts(normalize=True) * 100
plt.bar(ts_type_counts.index, ts_type_counts.values)
plt.title('Distribution of Third Shot Types in Model DataFrame')
plt.xlabel('Third Shot Type')
plt.ylabel('Percentage')
for i, v in enumerate(ts_type_counts.values):
```

```
plt.text(i, v + 1, f"{v:.1f}%", ha='center')
plt.ylim(0, 100)
plt.show()
```



```
In [114... # Create a train test split.
# Import packages.
import pandas as pd
from sklearn.model_selection import train_test_split
# create features
# Drop target variables and unnecessary columns
features = model_df.drop(columns=['rally_id', 'w_team_id', 'ts_type'])

#Get dummies for categorical variables
features = pd.get_dummies(features, drop_first=True)
target = model_df['ts_type']
# Split into 80 % training 20% test # use train_test_split
features_train, features_test, target_train, target_test= train_test_split(features,
target, test_size=0.2, random_state=42)

from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
target_train = le.fit_transform(target_train)
target_test = le.transform(target_test)
```

```
In [115... import numpy as np

# Check for NaNs in features_train
print("NaNs in features_train:", np.isnan(features_train).sum().sum())
print("NaNs in features_test:", np.isnan(features_test).sum().sum())

# Impute if any NaNs exist
from sklearn.impute import SimpleImputer
imputer = SimpleImputer(strategy='median')
features_train = imputer.fit_transform(features_train)
features_test = imputer.transform(features_test)
```

NaNs in features_train: 0
NaNs in features_test: 0

```
In [103... from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score
import xgboost as xgb
from sklearn.neural_network import MLPClassifier

# Logistic Regression Grid Search
lr_params = {'C': [0.01, 0.1, 1, 10], 'solver': ['liblinear', 'lbfgs']}
lr = LogisticRegression(max_iter=1000)
lr_grid = GridSearchCV(lr, lr_params, cv=3, scoring='accuracy')
lr_grid.fit(features_train, target_train)
lr_best = lr_grid.best_estimator_
lr_pred = lr_best.predict(features_test)
lr_acc = accuracy_score(target_test, lr_pred)
print(f"Best Logistic Regression Accuracy: {lr_acc:.3f}")

# Random Forest Grid Search
rf_params = {'n_estimators': [50, 100], 'max_depth': [None, 5, 10]}
rf = RandomForestClassifier(random_state=42)
rf_grid = GridSearchCV(rf, rf_params, cv=3, scoring='accuracy')
rf_grid.fit(features_train, target_train)
rf_best = rf_grid.best_estimator_
rf_pred = rf_best.predict(features_test)
rf_acc = accuracy_score(target_test, rf_pred)
print(f"Best Random Forest Accuracy: {rf_acc:.3f}")

# XGBoost Grid Search
xgb_params = {
    'n_estimators': [50, 100],
    'max_depth': [3, 6, 10],
    'learning_rate': [0.01, 0.1, 0.2]
}
xgb_clf = xgb.XGBClassifier(use_label_encoder=False, eval_metric='logloss', random_state=42)
xgb_grid = GridSearchCV(xgb_clf, xgb_params, cv=3, scoring='accuracy')
xgb_grid.fit(features_train, target_train)
xgb_best = xgb_grid.best_estimator_
xgb_pred = xgb_best.predict(features_test)
xgb_acc = accuracy_score(target_test, xgb_pred)
print(f"Best XGBoost Accuracy: {xgb_acc:.3f}")

# MLPClassifier (Neural Network) Grid Search
mlp_params = {
    'hidden_layer_sizes': [(16,), (32,), (64,), (32, 16)],
    'activation': ['relu', 'tanh'],
    'solver': ['adam'],
```

```

    'max_iter': [200]
}
mlp = MLPClassifier(random_state=42)
mlp_grid = GridSearchCV(mlp, mlp_params, cv=3, scoring='accuracy')
mlp_grid.fit(features_train, target_train)
mlp_best = mlp_grid.best_estimator_
mlp_pred = mlp_best.predict(features_test)
mlp_acc = accuracy_score(target_test, mlp_pred)
print(f"Best MLPClassifier Accuracy: {mlp_acc:.3f}")

# Print best model
accuracies = {
    'Logistic Regression': lr_acc,
    'Random Forest': rf_acc,
    'XGBoost': xgb_acc,
    'MLPClassifier': mlp_acc
}
best_model = max(accuracies, key=accuracies.get)
print(f"Best model: {best_model} with accuracy {accuracies[best_model]:.3f}")

```

```

/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
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to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
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https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

```

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(

```



```

/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
Best Logistic Regression Accuracy: 0.677
Best Random Forest Accuracy: 0.677

/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:07:26] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:07:29] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:07:32] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:07:35] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:07:41] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:07:47] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:07:52] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:07:56] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:08:01] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:08:05] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:08:12] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:08:19] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:08:27] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:08:32] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:08:37] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:08:43] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:08:52] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:09:01] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:09:11] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:09:14] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:09:17] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

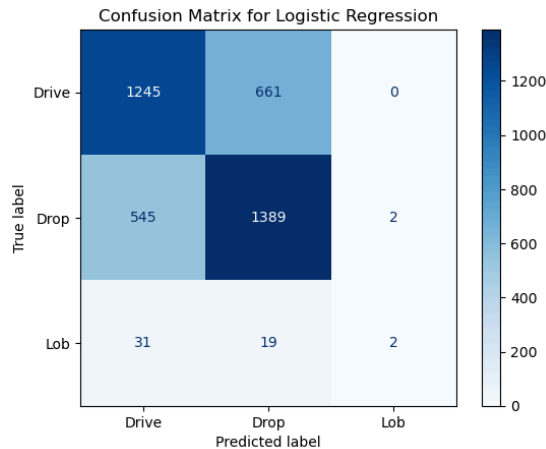
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [15:09:21] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

```

[illegible]

```
In [104]: # Show the confusion matrix for the best model
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
if best_model == 'Logistic Regression':
    best_pred = lr_pred
elif best_model == 'Random Forest':
    best_pred = rf_pred
elif best_model == 'XGBoost':
    best_pred = xgb_pred
elif best_model == 'MLPClassifier':
    best_pred = mlp_pred
cm = confusion_matrix(target_test, best_pred)
```

```
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=le.classes_)
disp.plot(cmap=plt.cm.Blues)
plt.title(f'Confusion Matrix for {best_model}')
plt.show()
```



```
In [105... # Show the feature importance for the best model if applicable
if best_model in ['Random Forest', 'XGBoost']:
    if best_model == 'Random Forest':
        importances = rf.best.feature_importances_
    elif best_model == 'XGBoost':
        importances = xgb.best.feature_importances_
    feature_names = features.columns
    feature_importance_df = pd.DataFrame({'Feature': feature_names, 'Importance': importances})
    feature_importance_df = feature_importance_df.sort_values(by='Importance', ascending=False)
    plt.figure(figsize=(10, 6))
    plt.barh(feature_importance_df['Feature'], feature_importance_df['Importance'])
    plt.xlabel('Importance')
    plt.title(f'Feature Importance for {best_model}')
    plt.gca().invert_yaxis()
    plt.show()
```

```
In [116... from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score
import xgboost as xgb
from sklearn.neural_network import MLPClassifier

# Logistic Regression Grid Search
lr_params = {'C': [0.01, 0.1, 1, 10], 'solver': ['liblinear', 'lbfgs']}
lr = LogisticRegression(max_iter=1000)
lr_grid = GridSearchCV(lr, lr_params, cv=3, scoring='accuracy')
lr_grid.fit(features_train, target_train)
lr_best = lr_grid.best_estimator_
lr_pred = lr_best.predict(features_test)
lr_acc = accuracy_score(target_test, lr_pred)
print(f"Best Logistic Regression Accuracy: {lr_acc:.3f}")
```

```
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.
```



```

features = pd.get_dummies(features, drop_first=True)
target = model_df['ts_type']
# Split into 80 % training 20% test # use train_test_split
features_train, features_test, target_train, target_test= train_test_split(features,
target, test_size=0.2, random_state=42)

from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
target_train = le.fit_transform(target_train)
target_test = le.transform(target_test)

```

In [107...

```

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score
import xgboost as xgb
from sklearn.neural_network import MLPClassifier

# Logistic Regression Grid Search
lr_params = {'C': [0.01, 0.1, 1, 10], 'solver': ['liblinear', 'lbfgs']}
lr = LogisticRegression(max_iter=1000)
lr_grid = GridSearchCV(lr, lr_params, cv=3, scoring='accuracy')
lr_grid.fit(features_train, target_train)
lr_best = lr_grid.best_estimator_
lr_pred = lr_best.predict(features_test)
lr_acc = accuracy_score(target_test, lr_pred)
print(f"Best Logistic Regression Accuracy: {lr_acc:.3f}")

# Random Forest Grid Search
rf_params = {'n_estimators': [50, 100], 'max_depth': [None, 5, 10]}
rf = RandomForestClassifier(random_state=42)
rf_grid = GridSearchCV(rf, rf_params, cv=3, scoring='accuracy')
rf_grid.fit(features_train, target_train)
rf_best = rf_grid.best_estimator_
rf_pred = rf_best.predict(features_test)
rf_acc = accuracy_score(target_test, rf_pred)
print(f"Best Random Forest Accuracy: {rf_acc:.3f}")

# XGBoost Grid Search
xgb_params = {
    'n_estimators': [50, 100],
    'max_depth': [3, 6, 10],
    'learning_rate': [0.01, 0.1, 0.2]
}
xgb_clf = xgb.XGBClassifier(use_label_encoder=False, eval_metric='logloss', random_state=42)
xgb_grid = GridSearchCV(xgb_clf, xgb_params, cv=3, scoring='accuracy')
xgb_grid.fit(features_train, target_train)
xgb_best = xgb_grid.best_estimator_
xgb_pred = xgb_best.predict(features_test)
xgb_acc = accuracy_score(target_test, xgb_pred)
print(f"Best XGBoost Accuracy: {xgb_acc:.3f}")

# MLPClassifier (Neural Network) Grid Search
mlp_params = {
    'hidden_layer_sizes': [(16,), (32,), (64,), (32, 16)],
    'activation': ['relu', 'tanh'],
    'solver': ['adam'],
    'max_iter': [200]
}
mlp = MLPClassifier(random_state=42)
mlp_grid = GridSearchCV(mlp, mlp_params, cv=3, scoring='accuracy')
mlp_grid.fit(features_train, target_train)
mlp_best = mlp_grid.best_estimator_
mlp_pred = mlp_best.predict(features_test)
mlp_acc = accuracy_score(target_test, mlp_pred)
print(f"Best MLPClassifier Accuracy: {mlp_acc:.3f}")

# Print best model
accuracies = {
    'Logistic Regression': lr_acc,
    'Random Forest': rf_acc,
    'XGBoost': xgb_acc,
    'MLPClassifier': mlp_acc
}
best_model = max(accuracies, key=accuracies.get)
print(f"Best model: {best_model} with accuracy {accuracies[best_model]:.3f}")

```

```

/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```

n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
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STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

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<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
Best Logistic Regression Accuracy: 0.531
Best Random Forest Accuracy: 0.568
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:24:17] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:24:17] WARNING: /Users/runne
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Parameters: { "use_label_encoder" } are not used.
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bst.update(dtrain, iteration=i, fobj=obj)
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r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

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bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:24:17] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:24:18] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:24:18] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:24:18] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

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Parameters: { "use_label_encoder" } are not used.
```

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bst.update(dtrain, iteration=i, fobj=obj)
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r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:24:19] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:24:19] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

[illegible]

Build simpler model with less features for amateur implementation.

```
In [108]: # Train again moving the location Date and the htm leaf
```

```

# If you again moving the location data into the mlc_obj
# ip players switched variables as this may be alot to process for a player in real time.

# Create a train test split.
# Import packages.
import pandas as pd
from sklearn.model_selection import train_test_split
# create features
# Drop target variables and unnecessary columns
features = model_df.drop(columns=['ts_type', 'rally_id', 'w_team_id', 'srv_team_rs_player_id',
                                'srv_team_ls_player_id', 'rtrn_team_rs_player_id',
                                'rtrn_team_ls_player_id', 'srv_team_id', 'srv_player_id', 'rtrn_team_id', 'rtrn_player_id', 'ts_player_id',
                                'btt_before', 'srv_switch_ind', 'rtrn_switch_ind', 'srv_team_flipped_ind',
                                'rtrn_team_flipped_ind'])

#Get dummies for categorical variables
features = pd.get_dummies(features, drop_first=True)
target = model_df['ts_type']
# Split into 80 % training 20% test # use train_test_split
features_train, features_test, target_train, target_test= train_test_split(features,
target, test_size=0.2, random_state=42)

from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
target_train = le.fit_transform(target_train)
target_test = le.transform(target_test)

```

In [109..

```

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score
import xgboost as xgb
from sklearn.neural_network import MLPClassifier

# Logistic Regression Grid Search
lr_params = {'C': [0.01, 0.1, 1, 10], 'solver': ['liblinear', 'lbfgs']}
lr = LogisticRegression(max_iter=1000)
lr_grid = GridSearchCV(lr, lr_params, cv=3, scoring='accuracy')
lr_grid.fit(features_train, target_train)
lr_best = lr_grid.best_estimator_
lr_pred = lr_best.predict(features_test)
lr_acc = accuracy_score(target_test, lr_pred)
print(f"Best Logistic Regression Accuracy: {lr_acc:.3f}")

# Random Forest Grid Search
rf_params = {'n_estimators': [50, 100], 'max_depth': [None, 5, 10]}
rf = RandomForestClassifier(random_state=42)
rf_grid = GridSearchCV(rf, rf_params, cv=3, scoring='accuracy')
rf_grid.fit(features_train, target_train)
rf_best = rf_grid.best_estimator_
rf_pred = rf_best.predict(features_test)
rf_acc = accuracy_score(target_test, rf_pred)
print(f"Best Random Forest Accuracy: {rf_acc:.3f}")

# XGBoost Grid Search
xgb_params = {
    'n_estimators': [50, 100],
    'max_depth': [3, 6, 10],
    'learning_rate': [0.01, 0.1, 0.2]
}
xgb_clf = xgb.XGBClassifier(use_label_encoder=False, eval_metric='logloss', random_state=42)
xgb_grid = GridSearchCV(xgb_clf, xgb_params, cv=3, scoring='accuracy')
xgb_grid.fit(features_train, target_train)
xgb_best = xgb_grid.best_estimator_
xgb_pred = xgb_best.predict(features_test)
xgb_acc = accuracy_score(target_test, xgb_pred)
print(f"Best XGBoost Accuracy: {xgb_acc:.3f}")

# MLPClassifier (Neural Network) Grid Search
mlp_params = {
    'hidden_layer_sizes': [(16,), (32,), (64,), (32, 16)],
    'activation': ['relu', 'tanh'],
    'solver': ['adam'],
    'max_iter': [200]
}
mlp = MLPClassifier(random_state=42)
mlp_grid = GridSearchCV(mlp, mlp_params, cv=3, scoring='accuracy')
mlp_grid.fit(features_train, target_train)
mlp_best = mlp_grid.best_estimator_
mlp_pred = mlp_best.predict(features_test)
mlp_acc = accuracy_score(target_test, mlp_pred)
print(f"Best MLPClassifier Accuracy: {mlp_acc:.3f}")

# Print best model
accuracies = {
    'Logistic Regression': lr_acc,
    'Random Forest': rf_acc,
    'XGBoost': xgb_acc,
    'MLPClassifier': mlp_acc
}
best_model = max(accuracies, key=accuracies.get)
print(f"Best model: {best_model} with accuracy {accuracies[best_model]:.3f}")

```

```

/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

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```

n_iter_i = _check_optimize_result(
/opt/anaconda3/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed
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Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
Best Logistic Regression Accuracy: 0.524
Best Random Forest Accuracy: 0.538
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:36] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:37] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
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/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:37] WARNING: /Users/runne
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Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:38] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:38] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:38] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:39] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:39] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
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r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:40] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:41] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:42] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:42] WARNING: /Users/runne
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bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:42] WARNING: /Users/runne
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Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:42] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

```



```
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:47] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:48] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:48] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
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/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:48] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:48] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:49] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
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```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:49] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

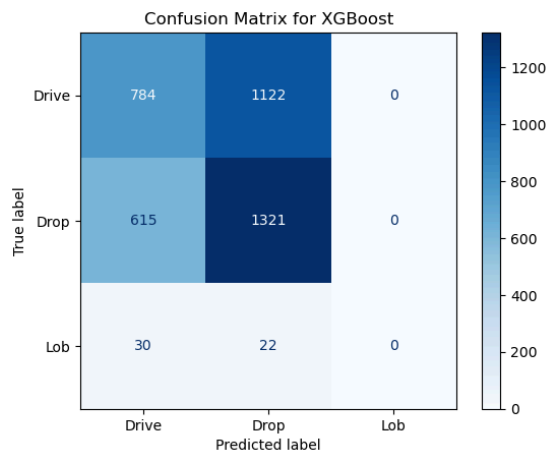
```
bst.update(dtrain, iteration=i, fobj=obj)
/opt/anaconda3/lib/python3.11/site-packages/xgboost/training.py:199: UserWarning: [16:25:49] WARNING: /Users/runne
r/work/xgboost/xgboost/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
Best XGBoost Accuracy: 0.541
Best MLPClassifier Accuracy: 0.535
Best model: XGBoost with accuracy 0.541
```

In []:

In [110..

```
# Show the confusion matrix for the best model
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
if best_model == 'Logistic Regression':
    best_pred = lr_pred
elif best_model == 'Random Forest':
    best_pred = rf_pred
elif best_model == 'XGBoost':
    best_pred = xgb_pred
elif best_model == 'MLPClassifier':
    best_pred = mlp_pred
cm = confusion_matrix(target_test, best_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=le.classes_)
disp.plot(cmap=plt.cm.Blues)
plt.title(f'Confusion Matrix for {best_model}')
plt.show()
```



In [111..

```
# Show the feature importance for the best model if applicable
if best_model in ['Random Forest', 'XGBoost']:
    if best_model == 'Random Forest':
        importances = rf_best.feature_importances_
    elif best_model == 'XGBoost':
        importances = xgb_best.feature_importances_
    feature_names = features.columns
    feature_importance_df = pd.DataFrame({'Feature': feature_names, 'Importance': importances})
    feature_importance_df = feature_importance_df.sort_values(by='Importance', ascending=False)
    plt.figure(figsize=(10, 6))
    plt.barh(feature_importance_df['Feature'], feature_importance_df['Importance'])
    plt.xlabel('Importance')
    plt.title(f'Feature Importance for {best_model}')
    plt.gca().invert_yaxis()
    plt.show()
```

